

- 1 (a) State *Archimedes Principle*.

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- (b) A man-made raft consists of a 1.00 m uniform plank weighing 0.20 kg attached to two hollow cubes, A and B, each of side 20 cm and of negligible weight. A 10.0 kg mass is placed on the centre of the plank and the raft floats on water with A and B submerged to depths of x and y respectively, as shown in Fig 1.1. The value of x and y are the same.

The density of water can be taken to be 1000 kg m^{-3} .

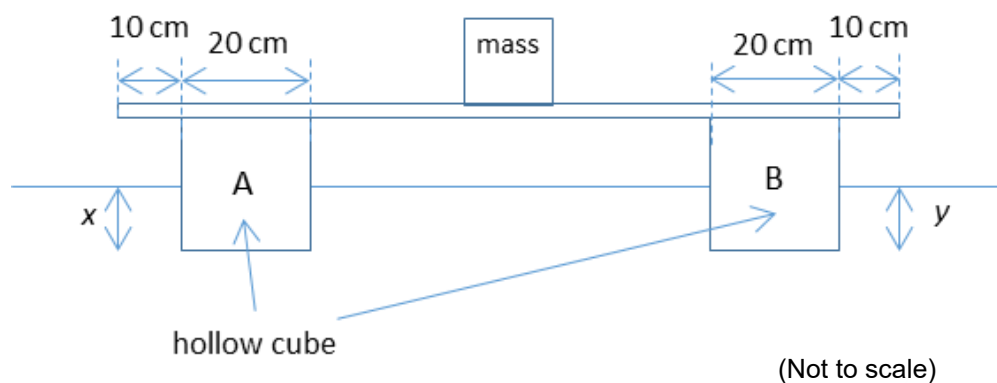


Fig 1.1

- (i) Determine x .

$x = \dots\dots\dots \text{ cm}$ [2]

- (ii) The mass is now moved by a small distance towards B and the raft remains afloat.

State and explain what will happen to x and y .

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- (iii) The mass is now placed directly above B and the raft tipped over and the mass dropped into the water.

Explain.

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